Multiply rational expressions

1

a $\frac{cf}{pv}$

b $\frac{35xy}{12}$

c $\frac{33rs}{35qt}$

d $\frac{77}{10y^2}$

e $\frac{16}{21}$

f $\frac{49}{40}$

g $3x$

h $-\frac{2ny^2}{9}$

i $\frac{14a}{33b}$

j $15y^6$

k $15m^2q$

l $20n^7$

2

a $\frac{9}{40y}$

b $\frac{1}{12y}$

c $\frac{5p-2}{5(p+7)}$

d $\frac{24x}{x-7}$

e $\frac{2(b-1)}{3}$

f $\frac{2(m-3)}{3}$

g $\frac{a+4}{4a}$

h $-\frac{8x}{x+8}$

i $-\frac{(x+2)(x-2)^2}{(x+6)(8-12x+6x^2-x^3)}$

j $\frac{9x(y-3)}{4y}$

Divide rational expressions

3

a $\frac{77xy}{24}$

b $\frac{2}{3}$

c $\frac{55}{36}$

d $\frac{15u}{16q}$

e $\frac{3}{2n}$

f $-\frac{8}{119v^2}$

g $\frac{2u}{3}$

h $\frac{5x^3z}{2y}$

i $\frac{uy^2}{tv^2}$

j $\frac{7y^5}{45x^3}$

4

a $\frac{2}{3}$

b $\frac{3}{k+1}$

c $\frac{1}{5(k-9)}$

d $\frac{15}{49}$

e $\frac{9m}{70y}$

f $\frac{1}{y}$

g $\frac{5(x+1)}{14}$

h $\frac{3(m+9)}{4(m+8)}$

i $\frac{5(x-6)}{3(x-3)(x+2)}$

j $\frac{x+y}{3}$

5

6

a Yes b No c No d Yes

7

a $\dfrac{\frac{v}{3}}{\frac{4u}{3}} = \dfrac{3\left(\frac{v}{3}\right)}{3\left(\frac{4u}{3}\right)} = \dfrac{v}{4u}$

b $\dfrac{8\left(\frac{u}{4}\right)}{8\left(\frac{v}{8}\right)} = \dfrac{2u}{v}$

c $\dfrac{\frac{y}{2}}{\frac{3x}{2}} = \dfrac{2\left(\frac{y}{2}\right)}{2\left(\frac{3x}{2}\right)} = \dfrac{y}{3x}$

d $\dfrac{n^2\left(\frac{4}{n}\right)}{n^2\left(\frac{5}{n^2}\right)} = \dfrac{4n}{5}$

8

a $\frac{2}{3}$ b $\frac{3}{2}$ c $3y$ d $-\frac{8x}{25}$

e $\frac{1}{3}$ f $\frac{1}{4}$ g $\frac{6}{x}$ h $\frac{4}{x}$

Applications

9 $\dfrac{9y}{5x}$

10 $\dfrac{5x^2y^3}{12q}$